

GED Mathematics: Geometry Practice Questions

MacLachlan College

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Practice Questions

1. Points, Lines, and Angles

1. Find the complement of a 42° angle.
2. Find the supplement of a 128° angle.
3. Two complementary angles are in ratio 3:2. Find both.
4. Two angles form a linear pair. One is 115° . Find the other.
5. If two vertical angles are $(3x + 10)^\circ$ and $(5x - 30)^\circ$, find x .

2. Triangles

6. Find the missing angle in a triangle with angles 47° and 83° .
7. A right triangle has legs 6 and 8. Find the hypotenuse.
8. Find the area of a triangle with base 9 and height 12.
9. The sides of a triangle are 5, 12, and 13. Determine if it is a right triangle.
10. An isosceles triangle has equal sides 10 and base 12. Find its height.

3. Quadrilaterals

11. Find the area of a rectangle with $l = 15$ and $w = 4$.
12. Find the area of a parallelogram with base 14 and height 8.
13. A trapezoid has bases 10 and 16 and height 6. Find its area.
14. A square has diagonal 12. Find its side length.
15. A rhombus has diagonals 10 and 24. Find its area.

4. Circles

16. Find the circumference of a circle with radius 9.
17. Find the area of a circle with diameter 10.
18. The circumference of a circle is 62.8. Find its radius.
19. A circle has area 113.04. Find its radius.
20. Find the area of a semicircle with diameter 8.

Solutions

1. Points, Lines, and Angles

1. $90 - 42 = 48^\circ$.
2. $180 - 128 = 52^\circ$.
3. $3x + 2x = 90 \Rightarrow 5x = 90 \Rightarrow x = 18$. Angles: 54° and 36° .
4. $180 - 115 = 65^\circ$.
5. $3x + 10 = 5x - 30 \Rightarrow 40 = 2x \Rightarrow x = 20$.

2. Triangles

6. $180 - (47 + 83) = 50^\circ$.
7. $c = \sqrt{6^2 + 8^2} = 10$.
8. $A = \frac{1}{2}(9)(12) = 54$.
9. $5^2 + 12^2 = 13^2 \Rightarrow 25 + 144 = 169$. True, it is right.
10. Height $= \sqrt{10^2 - 6^2} = 8$.

3. Quadrilaterals

11. $A = 15 \times 4 = 60$.
12. $A = 14 \times 8 = 112$.
13. $A = \frac{1}{2}(10 + 16)(6) = 78$.
14. $s = \frac{12}{\sqrt{2}} = 6\sqrt{2}$.
15. $A = \frac{1}{2}(10)(24) = 120$.

4. Circles

16. $C = 2\pi r = 18\pi$.
17. $A = \pi r^2 = \pi(5)^2 = 25\pi$.
18. $r = \frac{62.8}{2\pi} = 10$.
19. $r = \sqrt{\frac{113.04}{\pi}} = 6$.
20. $A = \frac{1}{2}\pi r^2 = \frac{1}{2}\pi(4)^2 = 8\pi$.